```
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
```

'wind\x6f\x77["\x6

\x65E\x6cemen\x74'

Hex
0x## - Numbers
\x## - Printable Chars

\x74

```
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
```

Hex

0x## - Numbers

\x## - Printable Chars

\x74

```
> [Convert]::ToInt16("74",16)
116
```

```
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
```

Hex

0x## - Numbers

\x## - Printable Chars

\x74

```
> [Convert]::ToInt16("74",16)
```

116

```
> [Char](116)
```

t

```
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
```

```
# Our hex encoded string
$cleanstring = $string =
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
```

```
# Our hex encoded string
$cleanstring = $string =
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'

# Use regex to pull out the literal \x## matches as a capture group
$hexChars   = $string | Select-String -Pattern "([.\\][.x].{2})" -AllMatches
```

```
# Our hex encoded string
$cleanstring = $string =
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'

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$hexChars   = $string | Select-String -Pattern "([.\\][.x].{2})" -AllMatches

# Loop through each match from our regex
foreach($char in $hexChars.Matches.Value){
```

```
# Our hex encoded string
$cleanstring = $string =
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# Loop through each match from our regex
foreach($char in $hexChars.Matches.Value){

   # Get just the value without the hex prefix
   $charVal    = $char.Substring(2,2)
```

```
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$cleanstring = $string =
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$hexChars   = $string | Select-String -Pattern "([.\\][.x].{2})" -AllMatches

# Loop through each match from our regex
foreach($char in $hexChars.Matches.Value){

   # Get just the value without the hex prefix
   $charVal    = $char.Substring(2,2)

   # Convert the value from base 16 to base 10
   $asciiVal   = [Convert]::ToInt16($charVal,16)
```

```
# Our hex encoded string
$cleanstring = $string =
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'

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   # Convert the value from base 16 to base 10
   $asciiVal   = [Convert]::ToInt16($charVal,16)

   # Convert the base 10 int to the ASCII char
   $asciiChar  = [Char]($asciiVal)
```

```
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$cleanstring = $string =
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   # Convert the base 10 int to the ASCII char
   $asciiChar  = [Char]($asciiVal)

   # Replace the hex char with the ASCII char
   $cleanstring = $cleanstring.Replace($char,$asciiChar)
}
```

```
# Our hex encoded string
$cleanstring = $string =
'wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'

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   # Convert the value from base 16 to base 10
   $asciiVal   = [Convert]::ToInt16($charVal,16)

   # Convert the base 10 int to the ASCII char
   $asciiChar  = [Char]($asciiVal)

   # Replace the hex char with the ASCII char
   $cleanstring = $cleanstring.Replace($char,$asciiChar)
}

> $cleanstring
window["document"].createElement
```

Can’t we just use AI?

ALWAYS LEADING

PowerShell

```
iex (Get-Secret -Name PSAI-gpt5 -AsPlainText);Set-OAIProvider AzureOpenAI;Set-AzOAISecrets @secrets
$s='wind\x6f\x77["\x64oc\x75\x6den\x74"\x5d\x2ec\x72\x65a\x74\x65E\x6cemen\x74'
Invoke-OAIChat "What is this string? $s"
Decoded (hex escapes → ASCII):

window["document"].createElement

This is JavaScript: it accesses the document object (window["document"] is the same as window.document) and references t
he createElement method (used to create DOM elements, e.g. document.createElement('div')).
```

Hackers Use PowerShell, Disable It!

But no, PowerShell is good!

We need to balance the risks & benefits

Probability

What is risk?

How do we manage risk?

Risk Management Frameworks (**RMF**)

GRC

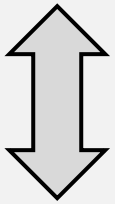
FRM

GRC: Governance, Risk, Compliance
FRM: Financial Risk Management

How do we manage risk?

Risk Management Frameworks (**RMF**)

GRC



FRM

ISO – General

COSO – Financial Centric

ISO: International Organization for Standardization
COSO: Committee of Sponsoring Organizations
GRC: Governance, Risk, Compliance
FRM: Financial Risk Management

How do we manage risk?

Risk Management Frameworks (**RMF**)

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How do we manage risk?

GRC

FRM

Risk Management Frameworks (**RMF**)

NIST/ITIL/COBIT – IT Centric

FAIR – Quantitative Focus

ISO – General

COSO – Financial Centric

NIST: National Institute of Standards and Technology
ITIL: Information Technology Infrastructure Library
COBIT: Control Objectives for Information and Related Technologies
FAIR: Factor Analysis of Information Risk
ISO: International Organization for Standardization
COSO: Committee of Sponsoring Organizations
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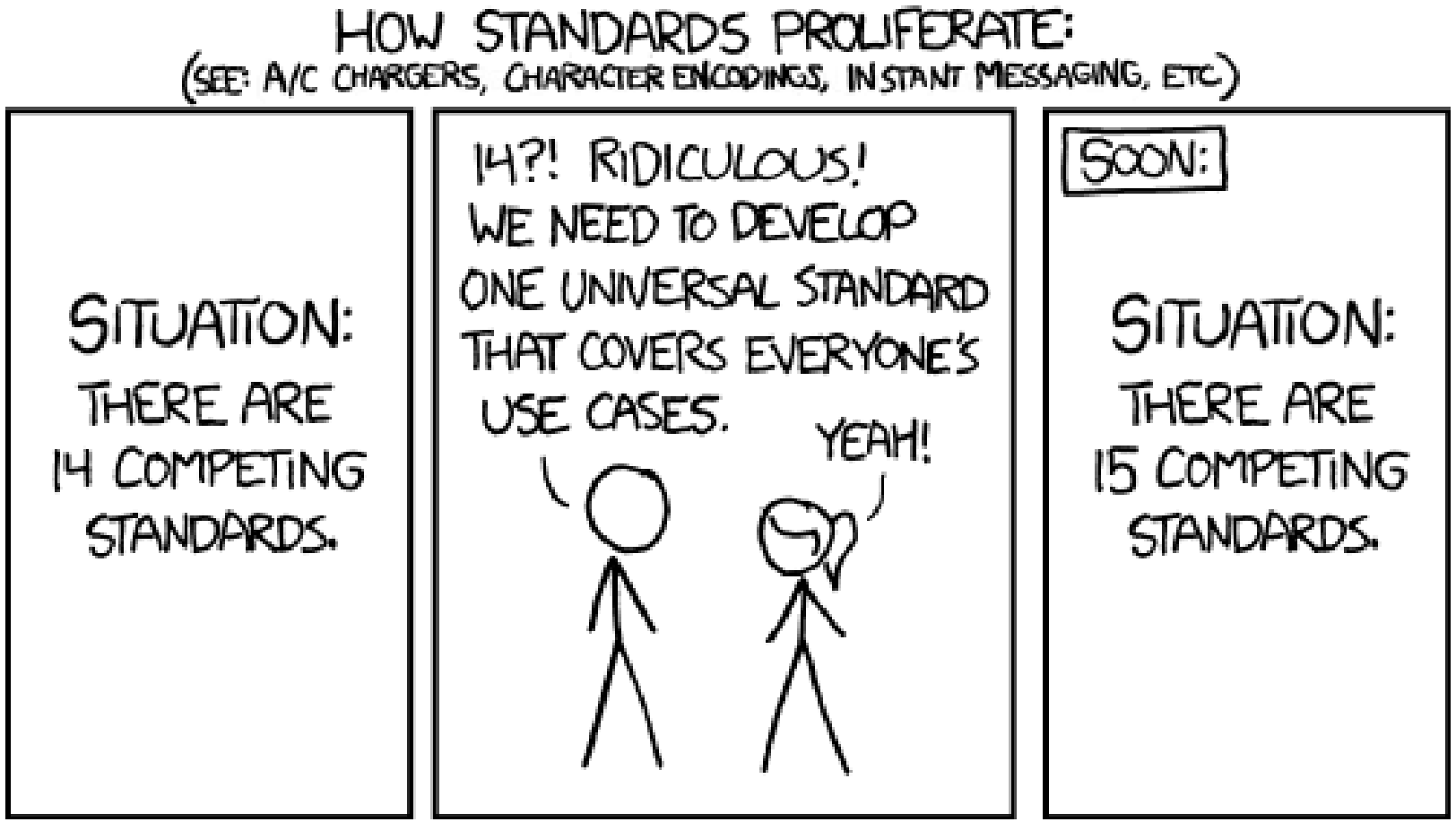
Fortunately, the charging one has been solved now that we've all standardized on mini-USB. Or is it micro-USB? Sh*t.



How do we manage risk?

Typically, through Security or Legal

Typically, IT is an afterthought

GRC

FRM

Risk Management Frameworks (**RMF**)

NIST/ITIL/COBIT – IT Centric

FAIR – Quantitative Focus

ISO – General

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NIST: National Institute of Standards and Technology
ITIL: Information Technology Infrastructure Library
COBIT: Control Objectives for Information and Related Technologies
FAIR: Factor Analysis of Information Risk
ISO: International Organization for Standardization
COSO: Committee of Sponsoring Organizations
GRC: Governance, Risk, Compliance
FRM: Financial Risk Management

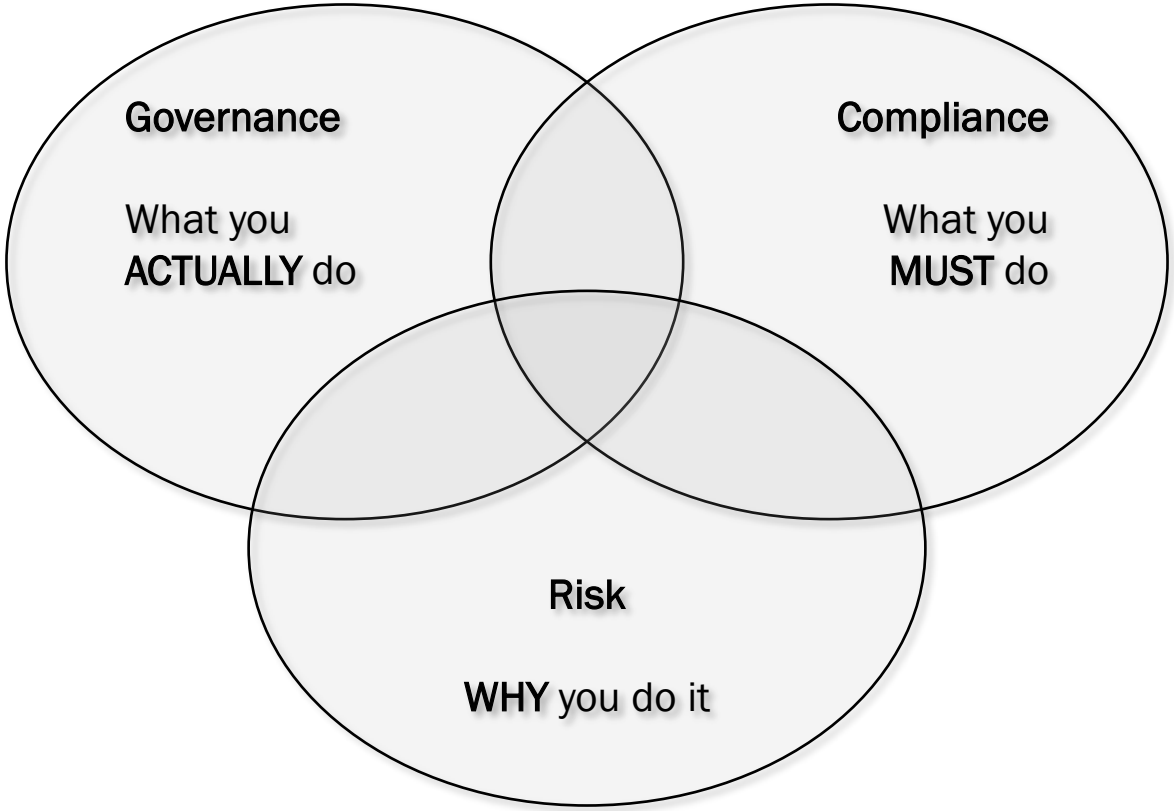
What is GRC then?

Risk

WHY you do it

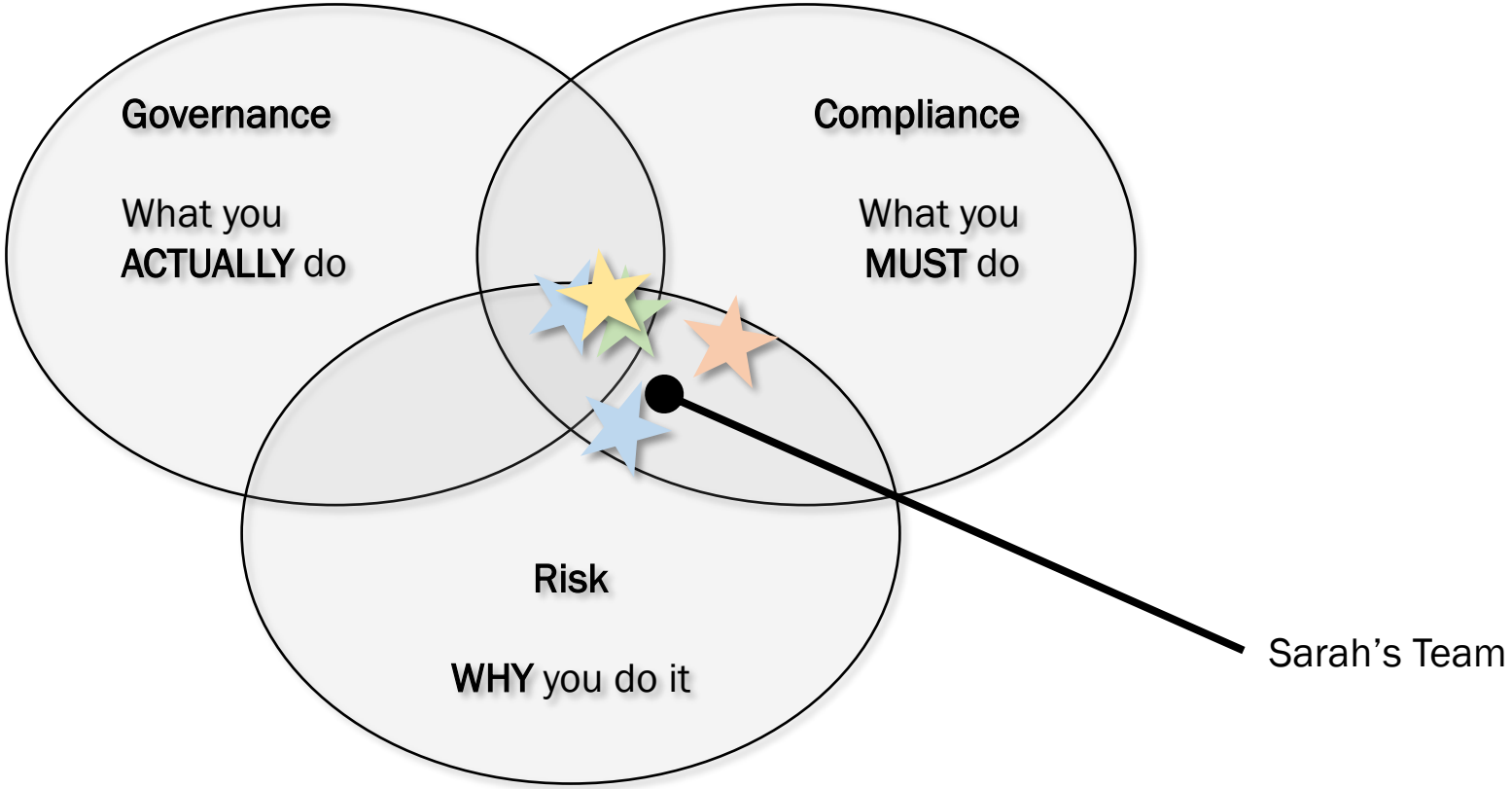
What is GRC then?

What is GRC then?

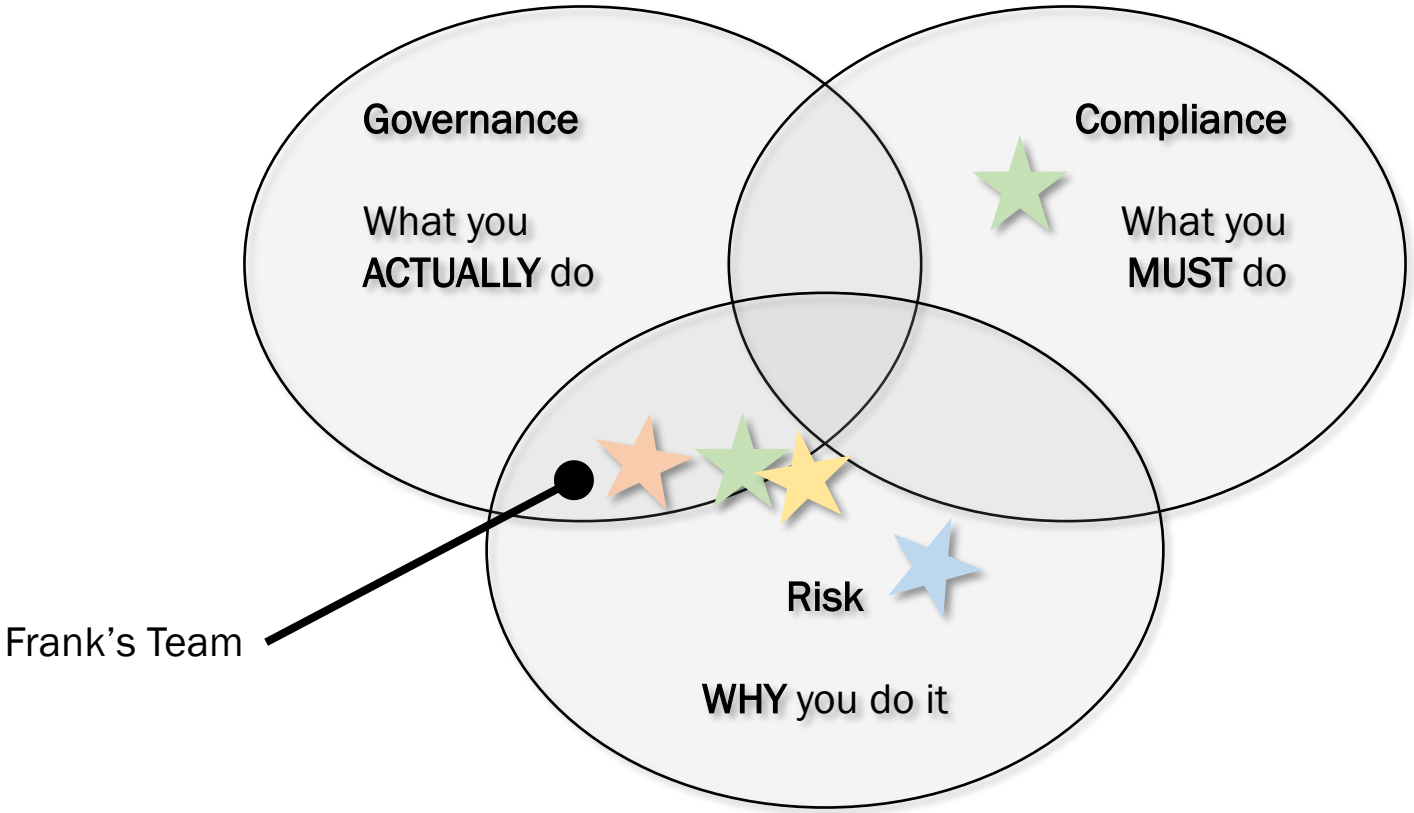


Unfortunately, it isn't a "We are here" representation

It is more like this…



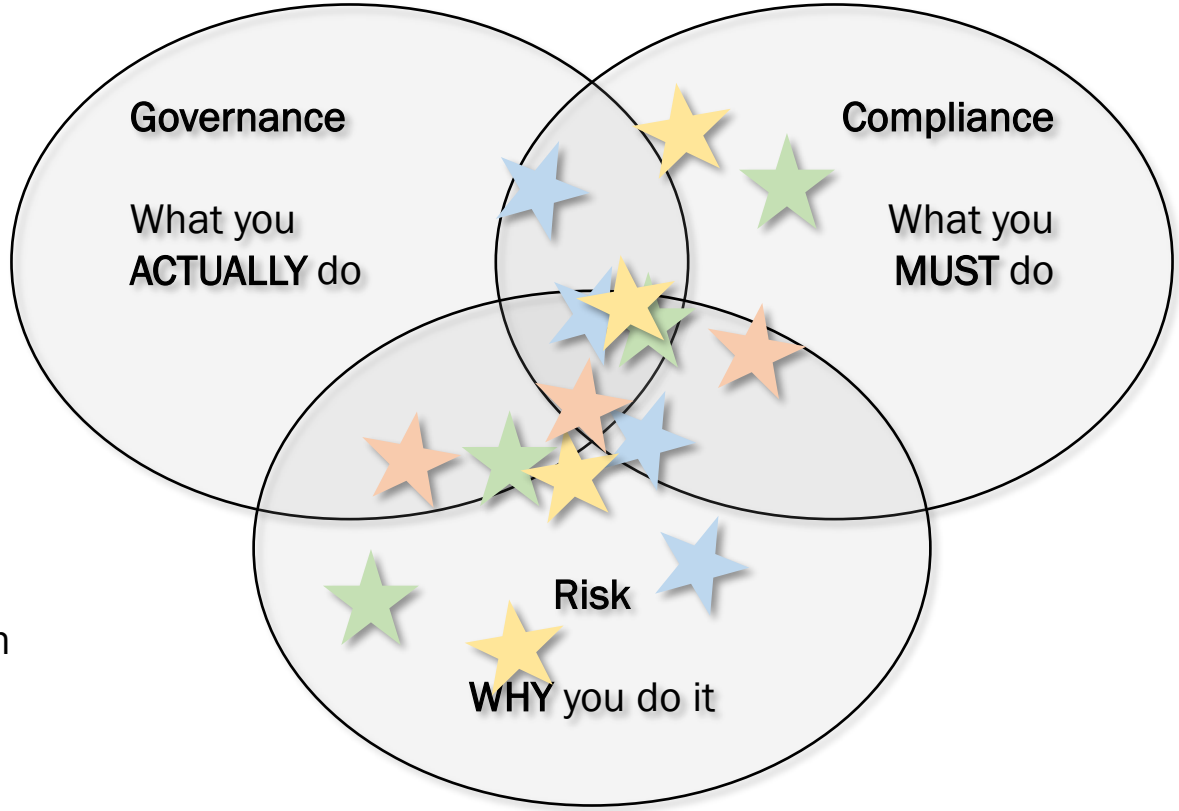
It is more like this…



It is more like this…

It is more like this…

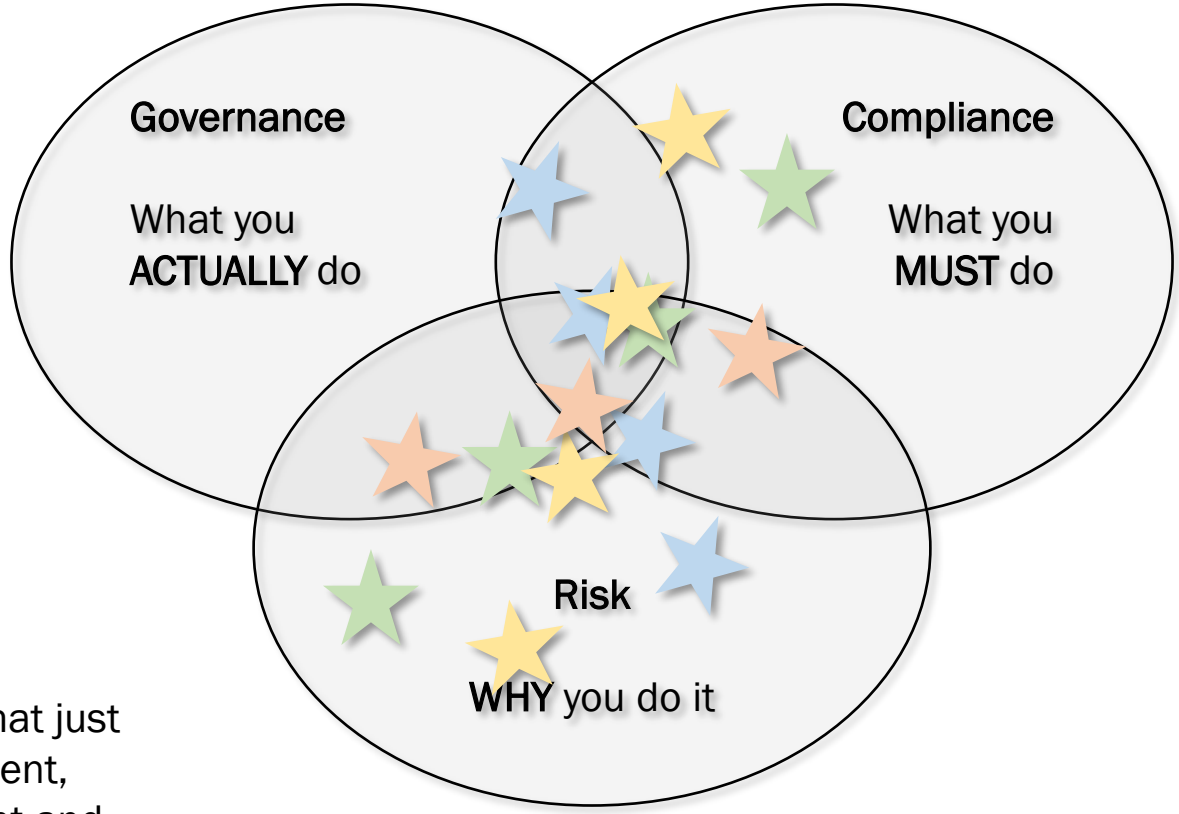
Mike's Team



Frank's Team

Sarah's Team

It is more like this…



Some random group that just bypassed procurement, bought new equipment and software, and is mining Bitcoin to fund their projects

How do we actually succeed then?

How do we actually succeed then?

ALWAYS LEADING

COMPLIANCE FORGE

Influencers (Internal & External)

Hierarchical cybersecurity & data protection governance starts with external influencers – these establish what is considered necessary for due diligence and due care for secure and compliant operations. These include statutory requirements (laws), regulatory requirements (government regulations) and contractual requirements (legally-binding obligations) that organizations must address.

External influencers usually impose meaningful penalties for non-compliance (e.g., HIPAA, NIST 800-171, FedRAMP, NY DFS 23 NYCRR Part 500, etc.). External influencers are often non-negotiable and are the primary source for defining a need for a policy and provide scoping for control objectives.

Internal influencers focus on management's desire for consistent, efficient and effective operations. This generally takes the form of:
- Business strategy
- Goals & objectives (e.g., customer satisfaction / service levels, budget constraints, quality targets, etc.)

Policies

Policies are high-level statements of management intent from an organization's executive leadership that are designed to influence decisions and guide the organization to achieve the desired outcomes.

Policies are enforced by Standards and further implemented by Procedures to establish actionable and accountable requirements.

Policies are a business decision, not a technical one. Technology determines how policies are implemented. Policies usually exist to satisfy an external requirement (e.g., law, regulation and/or contract).

Control Objectives

Control Objectives are targets or desired conditions to be met. These are statements describing what is to be achieved as a result of the organization implementing a Control.

Control Objectives directly link to laws, regulations and frameworks to align cybersecurity and data privacy with reasonably-expected practices. The intent is to establish sufficient evidence of due diligence and due care to withstand scrutiny (e.g., external audits/assessments) to disprove potential accusations of negligence.

Standards

Standards are mandatory requirements in regard to processes, actions, and configurations that are designed to satisfy Controls & Control Objectives.

Standards are intended to be granular and prescriptive to establish **Minimum Security Requirements (MSR)** that ensure systems, applications and processes are designed and operated to include appropriate cybersecurity and data privacy protections.

Guidelines

Guidelines are recommended practices that are based on industry-recognized secure practices. Guidelines help augment Standards when discretion is permissible.

Unlike Standards, Guidelines allow individuals / teams to apply discretion or leeway in interpretation, implementation, or use.

Controls

Controls are technical, administrative or physical safeguards. Controls are the nexus used to manage risks through preventing, detecting or lessening the ability of a particular threat from negatively impacting business processes.

Controls directly map to Standards, Procedures and Control Objectives. Control testing is designed to measure specific aspects of how Standards are actually implemented and if the Control / Control Objective is sufficiently addressed.

Assessment Objectives (AOs) are a set of determination statements that express the desired outcome for the assessment of a Control. AOs are the authoritative source of guidance for assessing controls to generate evidence to support the assertion that the underlying Control has been satisfied.

Procedures

Procedures are a documented set of steps necessary to perform a specific task or process in conformance with an applicable standard.

Procedures help address the question of how the organization actually operationalizes a Policy, Standard or Control. Without documented procedures, there will be no defensible evidence of due care practices.

Procedures are generally the responsibility of the process owner / asset custodian to build and maintain, but are expected to include stakeholder oversight to ensure applicable compliance requirements are addressed.

The result of a procedure is intended to satisfy a specific Control. Procedures are also commonly referred to as "control activities."

Risks

Risks represent a situation where someone or something valued is exposed to danger, harm or loss (noun) or to expose someone or something valued to danger, harm or loss (verb).

Risk is often calculated by a formula of the Occurrence Likelihood (OL) x the Impact Effect (IE) in an attempt to quantify the potential magnitude of a risk instance materializing.

In practical terms, a risk is associated with a Control deficiency (e.g., *if the control fails, what risk(s) is the organization exposed to?*)

While it is not possible to have a totally risk-free environment, it may be possible to manage risk by (1) avoiding, (2) reducing, (3) transferring, or (4) accepting risk(s).

An organization should maintain a "risk catalog" that contains organization-specific risks.

Threats

Threats represent a person or thing likely to cause damage or danger (noun) or to indicate impending damage or danger (verb).

In practical terms, a threat is a possible natural or man-made event that affects Control execution. (e.g., *if the threat materializes, will the control function as expected?*)

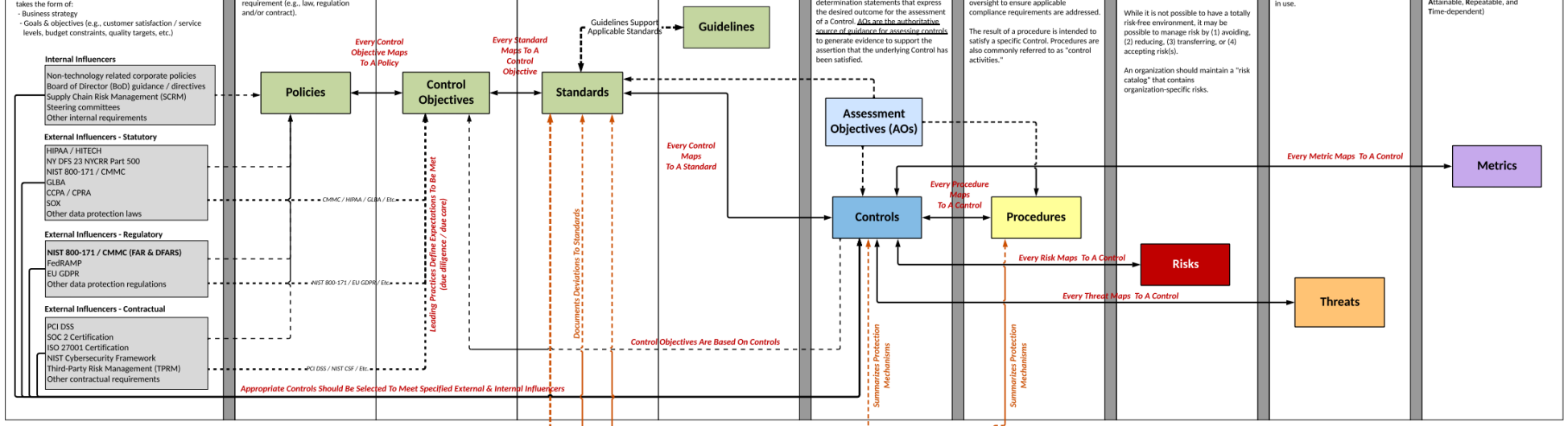
An organization should maintain a "threat catalog" that contains organization-specific natural and man-made threats that are applicable to its business operations and technologies in use.

Metrics

Metrics provide a "point in time" view of specific, discrete measurements, unlike trending and analytics that are derived by comparing a baseline of two or more measurements taken over a period of time. Analytics are generated from the analysis of metrics.

Analytics are designed to facilitate decision-making, evaluate performance and improve accountability through the collection, analysis and reporting of relevant performance-related data.

Good metrics are those that are **SMART** (**S**pecific, **M**easurable, **A**ttainable, **R**epeatable, and **T**ime-dependent)



SUPPORTING DOCUMENTATION

Secure Baseline Configurations

A set of specifications for a system, or Configuration Item (CI) within a system, application or service, that has been formally reviewed and agreed on at a given point in time, and which can be changed only through change control procedures. The secure baseline configuration is used as a basis for future builds, releases, and/or changes.

Risk Register / Plan of Action & Milestones (POA&M)

A POA&M is a document that identifies tasks that need to be accomplished (e.g., a formalized risk register). It details resources required to accomplish the elements of the plan, milestones for meeting the tasks and the scheduled completion dates for the milestones.

System Security Plan (SSP)

A SSP is a document that provides an overview of the security requirements for an information system and describes the security controls in place or planned for meeting those requirements.

Top-Down Process Flow of Cybersecurity & Privacy Governance Concepts

Internal & External Influencers primarily drive the development of cybersecurity and privacy policies. This requirements analysis is a component of governance, risk and compliance management practices to appropriately scope security program requirements.

Policies define high-level expectations and provide evidence of due diligence to address applicable requirements (internal and external).

Control Objectives support Policies and provide scoping for Standards, based on industry-recognized secure practices.

Standards operationalize Policies by providing organization-specific requirements that must be met.

Guidelines provide useful guidance that provides additional content to help operationalize Standards.

Controls are assigned to stakeholders to assign responsibilities in enforcing Standards.

Procedures operationalize Standards and Controls. The output of Procedures is evidence of due care to demonstrate that requirements are enforced.

Risks are associated with a control deficiency. (e.g., if the control fails, what risk is the organization exposed to?)

Natural and man-made threats affect control execution (e.g. if the threat materializes, will the control function as expected?)

Metrics provide evidence of an oversight function for the cybersecurity and privacy program by measuring criteria to determine performance.

How do we actually succeed then?

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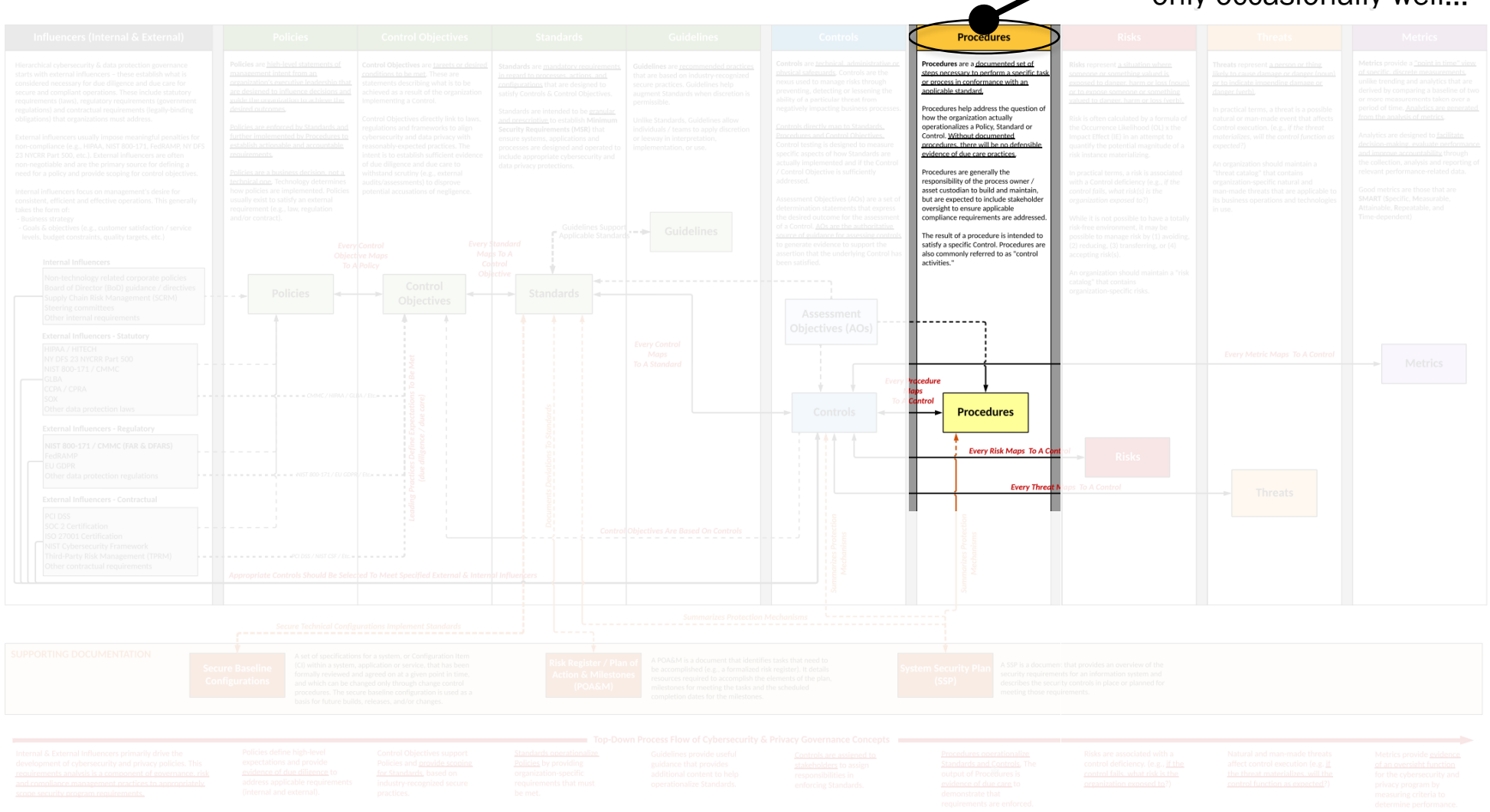
Most IT Ops teams do this, but only occasionally well…



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How do we actually succeed then?

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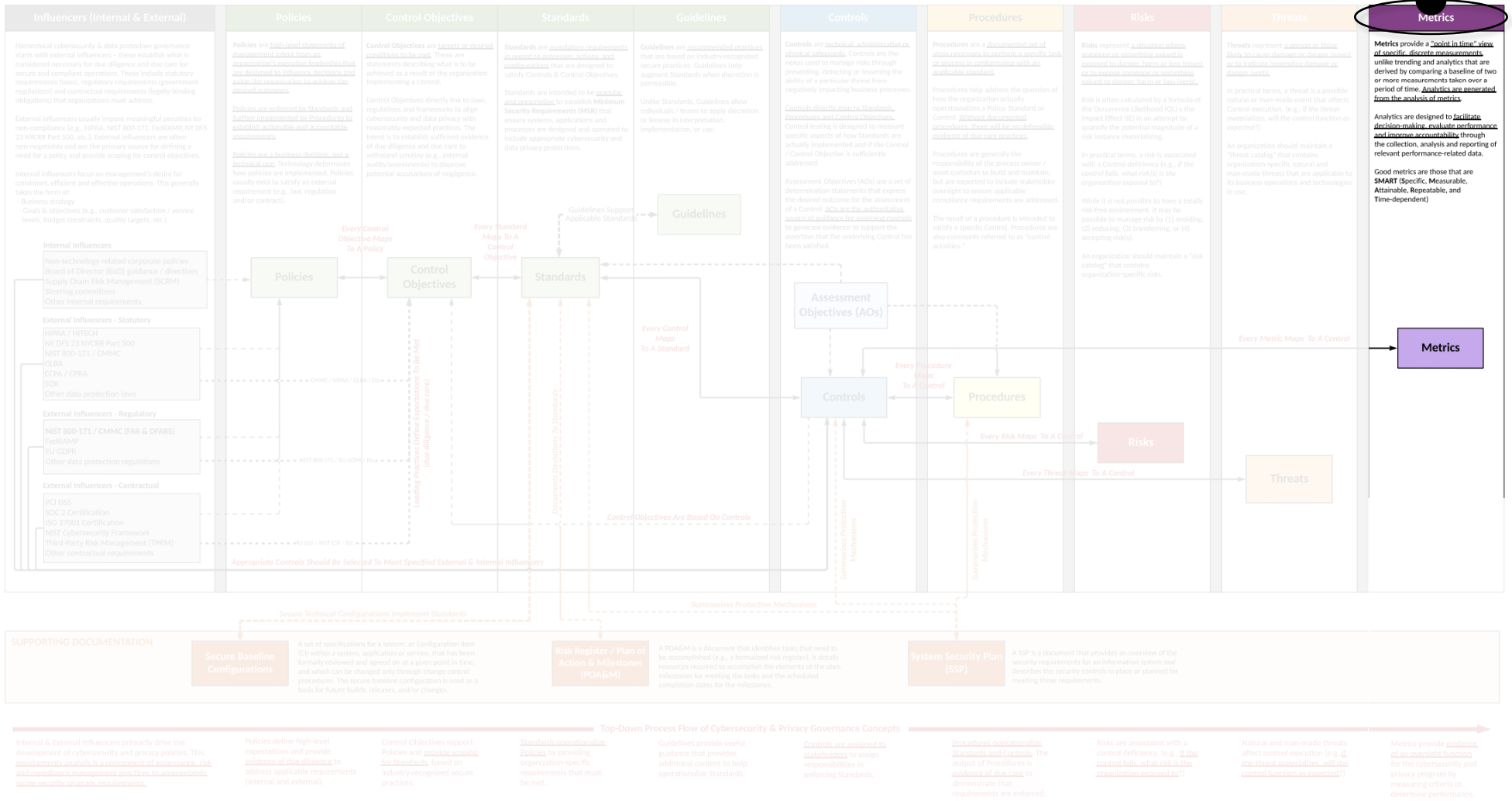
Sometimes you'll even be doing this...



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How do we actually succeed then?

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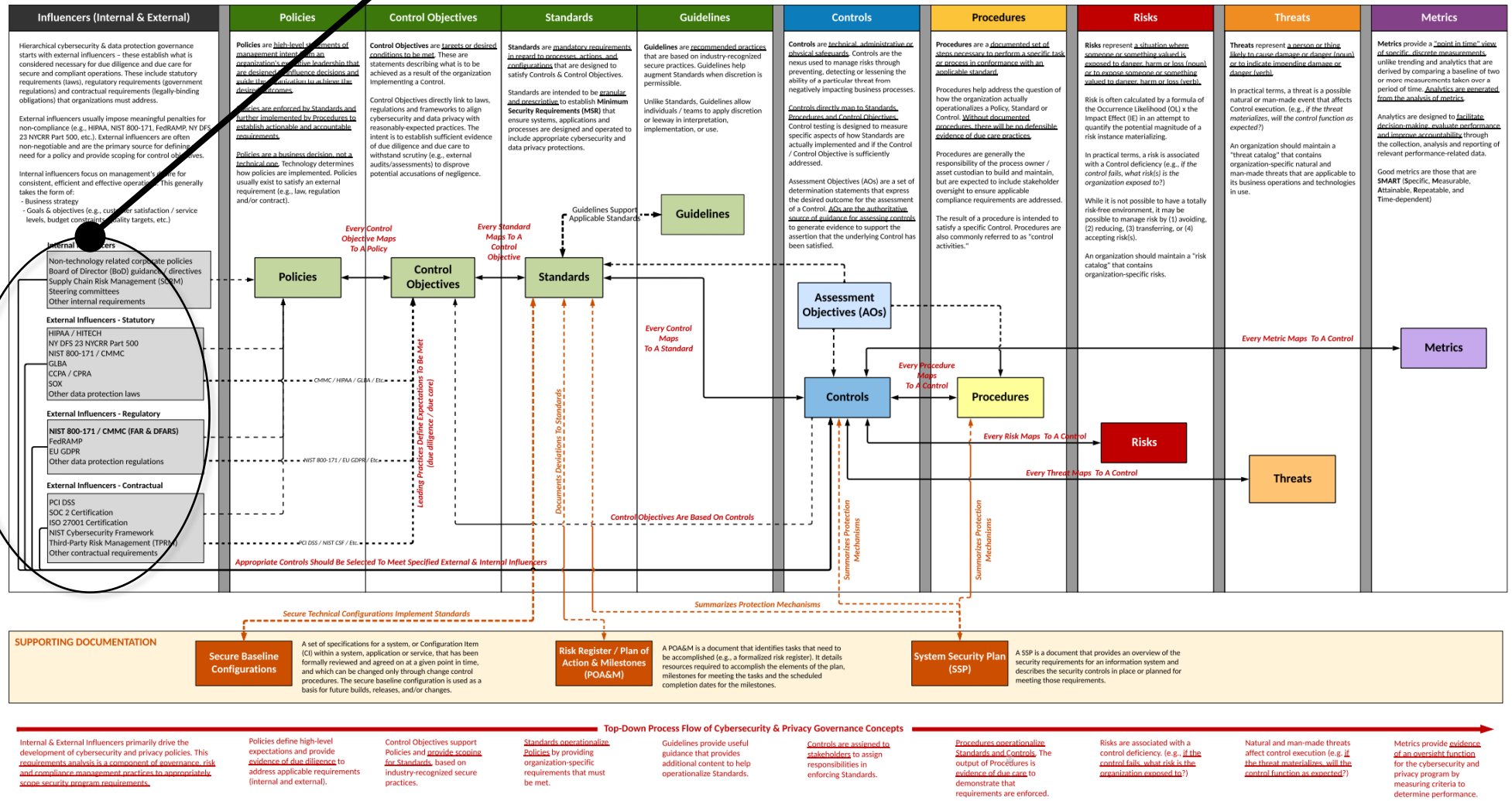
COMPLIANCE FORGE

Operations, Security, & Risk all need to align



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ALWAYS LEADING

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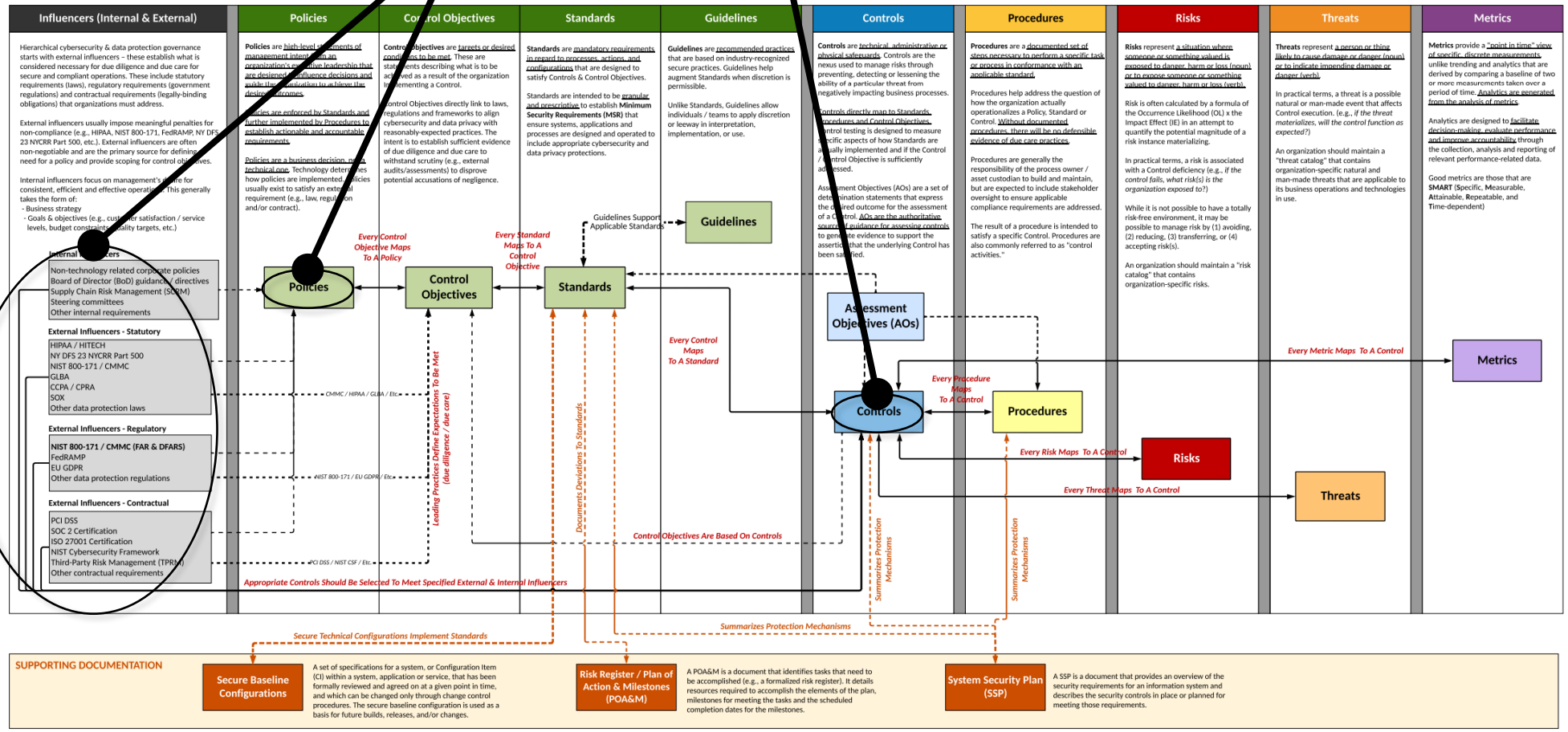

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How do we actually succeed then?

ALWAYS LEADING

COMPLIANCE FORGE

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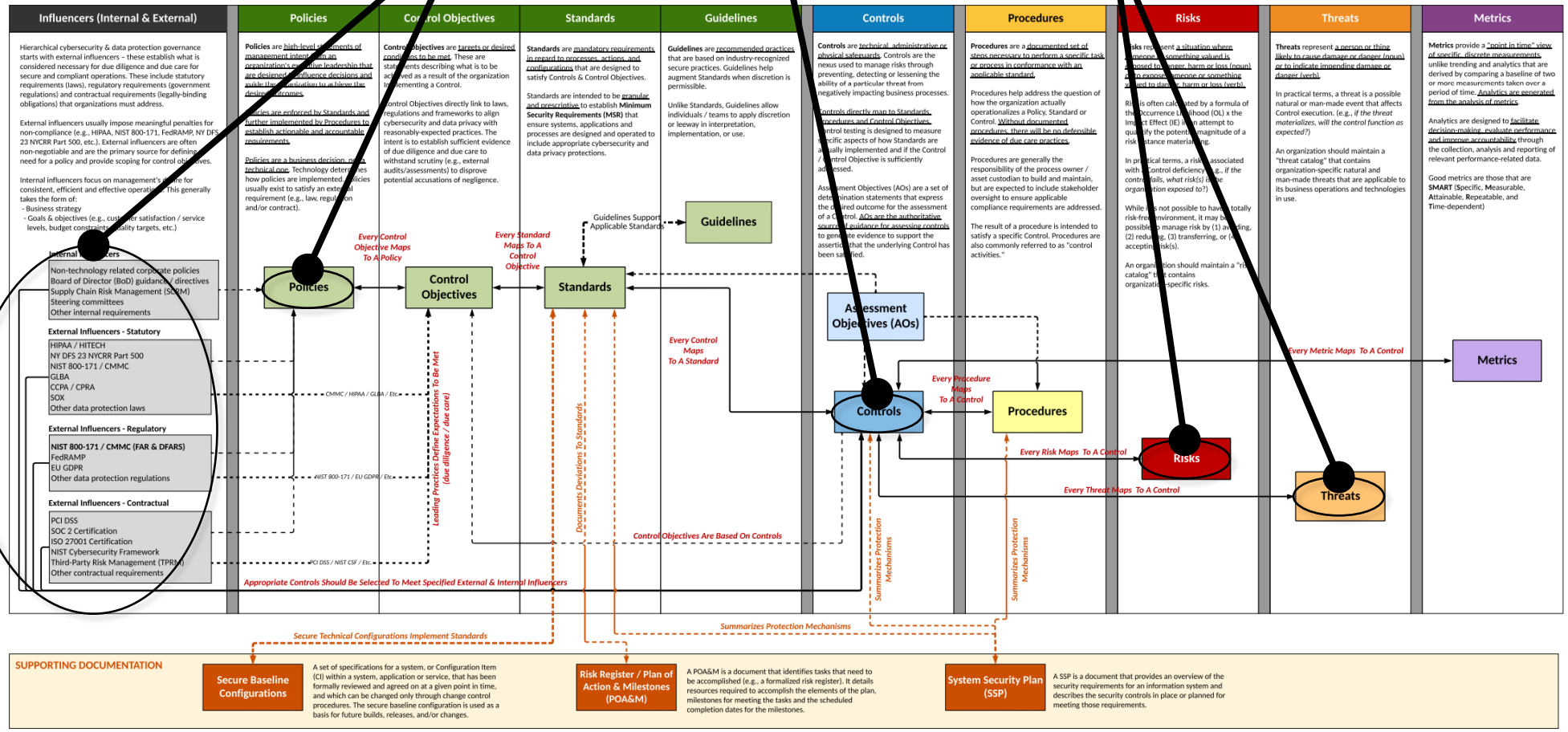

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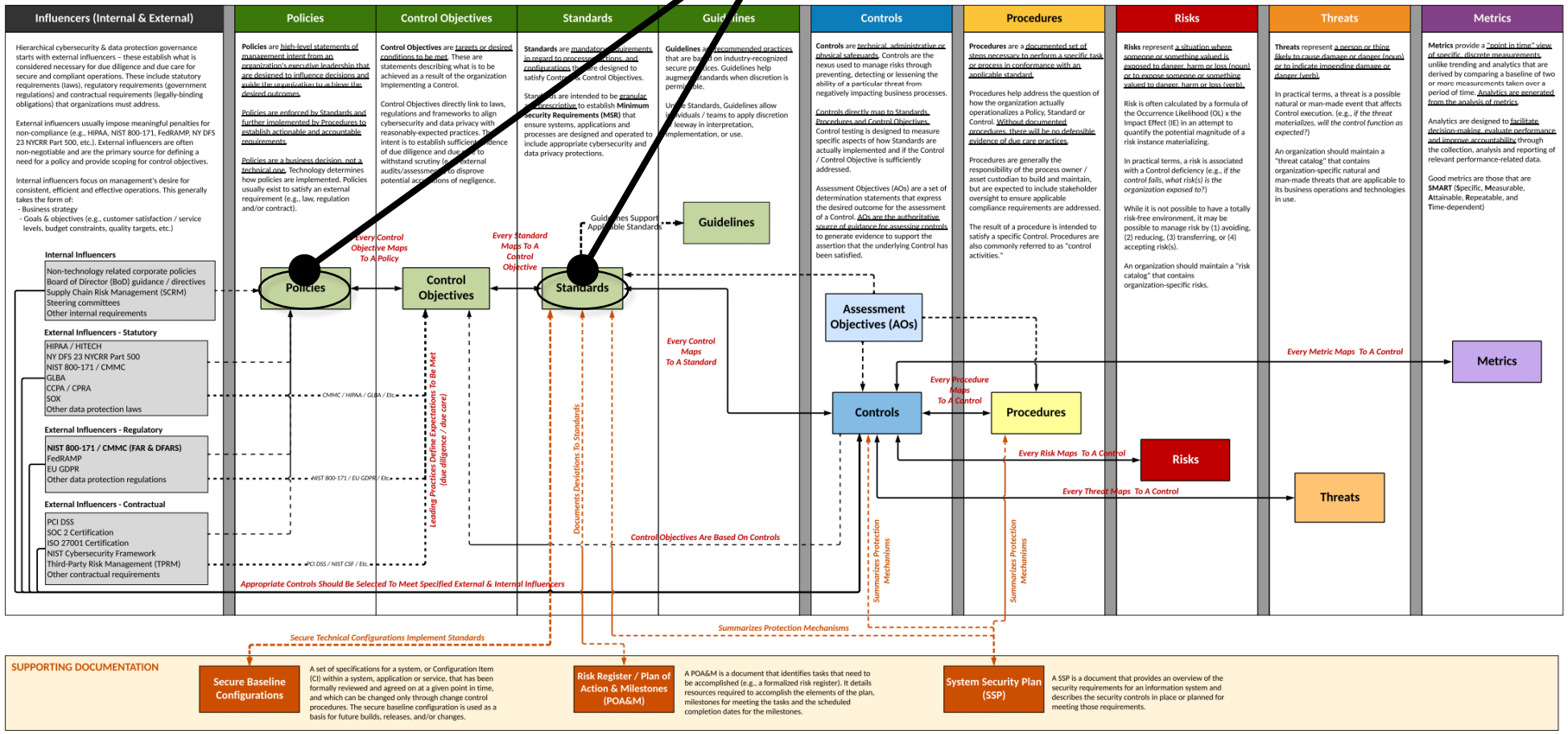
Quick opinion, your organization, likely, doesn't author standards…

ALWAYS LEADING

COMPLIANCE FORGE

Technical Standards **VS**
Risk Management Standards



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MITRE ATT&CK®

MITRE ATT&CK®

- **Command and Scripting Interpreter: PowerShell** (T1059.001)
  Adversaries may abuse PowerShell commands and scripts for execution.
  - Antivirus/Antimalware (M1049)
  - Code Signing (M1045)
  - Disable or Remove Feature or Program (M1042)
  - Execution Prevention (M1038)
  - Privileged Account Management (M1026)
  - Abuse of PowerShell for Arbitrary Execution (DET0455)

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  - Privileged Account Management (M1026)
  - Abuse of PowerShell for Arbitrary Execution (DET0455)
- **Event Triggered Execution: PowerShell Profile** (T1546.013)
  Adversaries may gain persistence and elevate privileges by executing malicious content triggered by PowerShell profiles.
  - Code Signing (M1045)
  - Restrict File and Directory Permissions (M1022)
  - Software Configuration (M1054)
  - Detection Strategy for PowerShell Profile Persistence via profile.ps1 Modification (DET0451)

How do I limit PowerShell risks?

- Anything Could Have Happened
- Anyone Can Run PowerShell
- Any Code Can Run
- Access from Anywhere to Anywhere
- Developers Make Mistakes

How do I limit PowerShell risks?

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    - Including Non-Human Identities
  - Software Restrictions (WDAC, AppLocker, SRP)
- Any Code Can Run

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  - Script Signing (Execution Policy)
  - Delegation Management (JEA)
- Access from Anywhere to Anywhere
  - Inbound (WinRM, SSH)
    - Including Authentication Challenges
  - Outbound (Host Firewall)
- Developers Make Mistakes

How do I limit PowerShell risks?

- Anything Could Have Happened
  - Audit Logging
  - Endpoint Detection & Response (EDR)
- Anyone Can Run PowerShell
  - Administrative Tiers (ESAE, RaMP)
    - Including Non-Human Identities
  - Software Restrictions (WDAC, AppLocker, SRP)
- Any Code Can Run
  - Script Signing (Execution Policy)
  - Delegation Management (JEA)
- Access from Anywhere to Anywhere
  - Inbound (WinRM, SSH)
    - Including Authentication Challenges
  - Outbound (Host Firewall)
- Developers Make Mistakes
  - Secret Management, AppSec Testing

Stack Ranking of Mitigating Control Topics

ALWAYS LEADING

Topic (MITRE ID)	Effort (High/Low)	Benefit (High/Low)	
Audit Logging (M1054)	Low	High	Existing Resources and Processes
EDR (M1049)	Low	High	
Remoting (M1054)	Low	High	
AppSec*	Low	Low	Requires New Resources or Processes
Script Signing* (M1045)	High	High	
Admin Tiering (M1026)	High	High	
Host Firewall	High	High	
Secret Management*	High	High	
Software Restrictions* (M1042)	High	High	
Delegation Management (M1038)	High	Low	

* All components of Software Development Life Cycle (SDLC)

Australian Government
Australian Signals Directorate

ASD AUSTRALIAN SIGNALS DIRECTORATE

ACSC Australian Cyber Security Centre

EN

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Securing PowerShell in the enterprise

https://www.cyber.gov.au/business-government/protecting-devices-systems/system-administration/securing-powershell-in-the-enterprise

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